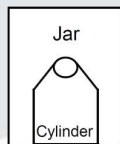
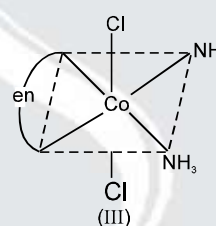
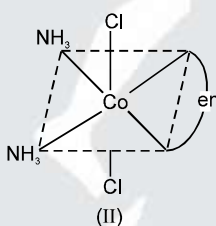
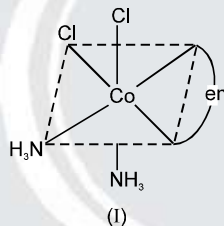


SCQ (Single Correct Type) :

- In context with the highly pure dilute solution of sodium in liquid ammonia, which of the following is the incorrect statement ?
 (A) In dilute solution the main species are metal ions (Na^+) and electrons, which are solvated (i.e. ammoniated)
 (B) It is paramagnetic, with approximately one unpaired electron per metal atom.
 (C) It acts as powerful oxidising agent.
 (D) It shows blue colour and exhibits electrical conductivity.
- A 20 litre rigid cylinder containing N_2 at 20 atm and 300 K is placed in big jar containing air at 1 atm and 300 K and finally sealed as given below. Now $\text{N}_2(\text{g})$ was allowed to leak into to jar at constant temperature. Final total pressure in jar is 11 atm. Final volume of jar in litre.



- (A) 18 litre (B) 40 litre (C) 58 litre (D) 38 litre
- Three arrangements are shown for the complex $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$. Pick up the correct statement.



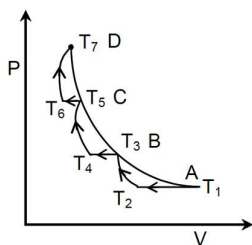
- (A) I and II are geometrical isomers (B) II and III are optical isomers
 (C) I and III are optical isomers (D) II and III are geometrical isomers

MCQ (Multiple Correct Type) :

- A hypothetical solid A^+B^- has $\frac{r_{\text{A}^+}}{R_{\text{B}^-}} = \frac{2 - \sqrt{2}}{\sqrt{2}}$ then correct statement(s) is/are :

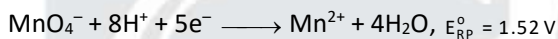
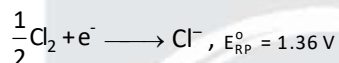
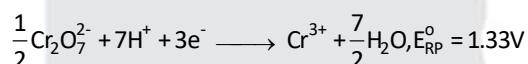
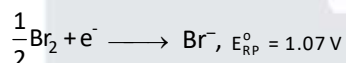
- (A) B^- touches total 18 ions
 (B) A^+ touches total 6 ions
 (C) Each body diagonal passes through the centre of 2 cations and 1 anion of a unit cell.
 (D) Each body diagonal passes through the centre of 2 anions and 1 cation of a unit cell.

5. Through a number of isobaric cooling and adiabatic compression steps 1 mol helium (consider as an ideal gas) is compressed from 1 bar, 300K to 32768 bar, 300K. Let the process be accomplished in three steps [Step AB, BC & CD]. At each step helium is firstly cooled and then compressed. At the end of each step the pressure increases 32 times and temperature returns to original value 300K as shown in diagram [R = 2 cal/K/mol]. Select correct option(s) :



- (A) $T_1 = T_3 = T_5 = T_7$ (B) $T_2 = T_4 = T_6$ (C) $Q_{\text{total}} = -3375 \text{ cal}$ (D) $DS_{\text{sys}} = 15 R \ln 2$
6. An aqueous solution of iodine become colorless on adding ?
- (A) Excess of $\text{Na}_2\text{S}_2\text{O}_3$ solution (B) Excess of Cl_2 water
(C) Excess of NaOH solution (D) Excess of NaBr solution

7. Some standard redox potentials are given below:



Which of the following statement(s) is/are correct under standard conditions ?

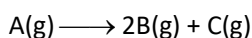
- (A) Bromine will oxidise chloride ions to chlorine
(B) Permanganate ions in acidic solution will oxidise chloride ions to chlorine
(C) Permanganate ions in acidic solution will oxidise chromium (III) ions to dichromate ions
(D) Dichromate ions in acidic solution will oxidise manganese (II) ion
8. Colemanite $\xrightarrow{\text{Na}_2\text{CO}_3}$ x $\xrightarrow{\text{HCl}}$ y $\xrightarrow{\Delta}$ z $\xrightarrow{\text{Mg}}$ w
Lewis acid an element

Identify incorrect statement(s) :

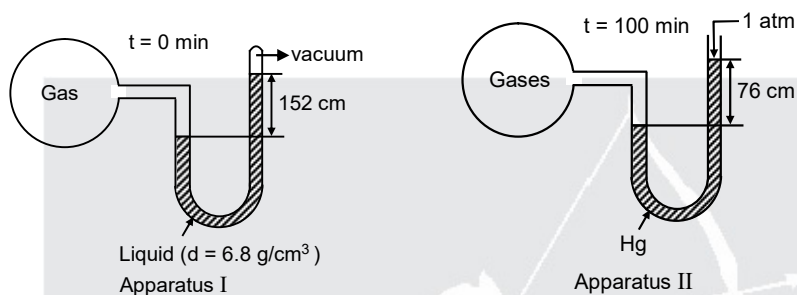
- (A) Oxidation number of boron in colemanite is +3.
(B) [z] when heated with Co gives coloured bead.
(C) Almost all compounds of [w] are ionic.
(D) Decahydrated [x] has 3, O-H bonds.
9. An element X has EAN = 36 for its $[\text{X}(\text{NH}_3)_4]^{2+}$ complex. Which of the following statement is /are correct for X?
- (A) X form amphoteric oxide

- (B) When phenol is heated with X, the product is benzene
 (C) It is a transition element belonging to 3d series
 (D) It is used in clemmensen reduction.

10. Given a first order reaction which takes place as follows :



Total pressure observed is as follows :



Which is/are correct statement(s) ? [Given $\ln 2 = 0.7$]

- (A) Rate constant of appearance of B is equal to $7 \times 10^{-3} \text{ min}^{-1}$
 (B) Half life period of reaction is equal to 100 min.
 (C) 75% of reaction is completed in 200 min.
 (D) 87.5% of reaction is completed in 100 min.
11. For the reaction $A \longrightarrow B$, the rate law expression is $-\frac{d[A]}{dt} = k[A]^{1/2}$. If initial concentration of A is A_0 , which of the following statement(s) is/are true regarding this reaction?
- (A) The plot of $\sqrt{A}$ against 't' will be linear.
 (B) The half life period is $t_{1/2} = \frac{\sqrt{2}(\sqrt{2} - 1)}{k} \sqrt{A_0}$
 (C) The rate of reaction decrease linearly with time.
 (D) The integrated form of rate expression is $\sqrt{A} = -\frac{k}{2}t + \sqrt{A_0}$
12. A salt mixture contain 0.018 mol each of NaI, NaBr and NaCl. When the mixture is dissolved in 200 mL of 0.24 M AgNO_3 , some precipitates are formed. Volume of final solution is 200 mL. Given (K_{sp} of AgI, AgBr and AgCl = 10^{-17} , 10^{-13} , 10^{-10} respectively). Which of the following is/are correct?
- (A) Nearly all I^- and Br^- are precipitated out.
 (B) $[\text{Ag}^+]$ in the final solution is $\frac{10^{-8}}{9} \text{ M}$
 (C) Total moles of silver salt in the precipitate is 0.036 mol.
 (D) Concentration of $[\text{I}^-]$ and $[\text{Br}^-]$ in the final solution is nearly $3 \times 10^{-9} \text{ M}$ and $3 \times 10^{-5} \text{ M}$ respectively.

13. For reaction : $\text{NH}_3 + \text{Na} \xrightarrow{\Delta} \text{P} + \text{Q}$
 Correct statement(s) is/are :
 (A) Hydrolysis of 'P' produces NH_3 gas.
 (B) Gas 'Q' reacts with Li to produce a reactive solid.
 (C) Anionic part of 'P' is weaker base than NH_3 .
 (D) Gas Q is also produced, when Li metal is heated with NH_3 gas.
14. A Daniel Cell is made at 25°C by connecting Zn/Zn^{+2} (0.1 M, 1 litre) & Cu/Cu^{+2} (0.9 M, 1 litre) electrodes. The cell is discharged when its emf reaches 1.10 volt, then it is charged (reversing the discharge process) by passing 0.6 F charge. Assume volume remains constant.
 [Given: $E^\circ_{\text{Zn}/\text{Zn}^{+2}} = 0.76\text{V}$, $E^\circ_{\text{Cu}/\text{Cu}^{+2}} = -0.34\text{V}$, $\frac{2.303RT}{F} = 0.06\text{V}$, $\log 2 = 0.30$]
 Select the correct option(s) :
 (A) Concentration of Zn^{+2} becomes 1 M when EMF reach 1.10 volt.
 (B) Concentration of Cu^{+2} becomes 0.5 M when EMF reach 1.10 volt.
 (C) EMF of the cell just after charging is 1.118 volt
 (D) Just after charging concentration of Cu^{+2} becomes 0.8 M
15. Given that molar conductivities at infinite dilution (25°C):
 $\text{HCl} = 426 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$
 $\text{NaCl} = 126 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$
 $\text{NaC (sodium crotonate)} = 83 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$
 The conductivity of a 0.001 M Crotonic acid solution is $3.83 \times 10^{-5} \Omega^{-1} \text{cm}^{-1}$
 Which of the following is/are correct?
 (A) $\lambda^\infty_{\text{M}}$ of crotonic acid is $383 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ (B) K_a of crotonic acid is 1.11×10^{-5}
 (C) K_a of crotonic acid is 1.11×10^{-7} (D) α of 0.01 M solution of crotonic acid is 0.1
16. On electrolysis, in which of the following, O_2 would be liberated at the anode ?
 (A) dilute H_2SO_4 with Pt electrodes.
 (B) aqueous AgNO_3 solution with Pt electrodes.
 (C) dilute H_2SO_4 with Cu electrodes.
 (D) aqueous NaOH solution with Fe cathode & Pt anode.

Match the column

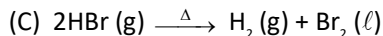
17. Match the following :

List – I

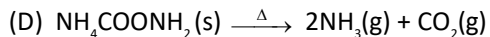
List – II

For 1 mole of reactant placed in open container in each reaction

- | | |
|--|---|
| (A) $2\text{NH}_3(\text{g}) \xrightarrow{\Delta} \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ | (P) The ratio of molecules of gaseous products is 1 : 1 |
| (B) $\text{PCl}_5(\text{g}) \xrightarrow{\Delta} \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ | (Q) 11.2 litre gaseous product at STP |



(R) mass of one of the product is 3 g



(S) 67.2 litre gaseous product at STP

18. Match the reactions given in List-I with the result of the respective reactions with appropriate options given in List-II.

	LIST-I		LIST-II
(I)	$P_4 + SOCl_2 \longrightarrow$	(P)	Preparation of diatomic halide
(II)	$KI + H_2O + O_3 \longrightarrow$	(Q)	Preparation of dioxygen
(III)	$SO_2 + O_2 \xrightarrow{V_2O_5/heat}$	(R)	Preparation of cyclic compound at room temperature
(IV)	$H_3PO_3 + Br_2 \xrightarrow{\text{heat in a sealed tube}}$	(S)	Preparation of acidic oxide
		(T)	Zero oxidation state of minimum two products
		(U)	In one of the products the central atom is in +5 oxidation state.

18. Which of the following has the correct combination considering List-I and List-II ?
 (A) (I - P) (B) (II - Q) (C) (III - R) (D) All of these
19. Which of the following has the incorrect combination considering List-I and List-II ?
 (A) (I - R) (B) (II - T) (C) (III - S) (D) (IV - U)

(Paragraph) :

20. How many of the following order(s) are incorrect with respect to property indicated ?

- (1) $Al^{3+} < Mg^{2+} < Li^+ < K^+$ - ionic size.
- (2) $B < C < N > O$ - ionization energy.
- (3) $F < Cl > Br < I$ - electron affinity.
- (4) $Si < P < C < N$ - electronegativity.
- (5) $N_2O_3 > CO_2 > B_2O_3 > BeO > Li_2O$ - acidic character.
- (6) $Na > Mg > Al > Si$ - metallic character.
- (7) $F_2 < Cl_2 < Br_2 < I_2$ - oxidising power.

21. In how many of following molecules, all atoms are not in same plane?

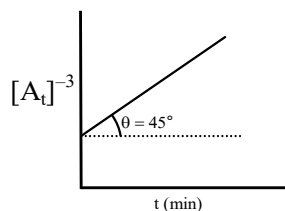
ClF_3	H_2O	PCl_3	BF_3
SF_4	H_2S	OCl_2	SO_3
XeF_6	NH_3	C_6H_6	XeF_2
XeF_4	PCl_5	I_2Cl_6	PH_3

22. The solubility of AgCN in a buffer solution is 1.58×10^{-5} mol litre⁻¹. Given K_{sp} of AgCN = 1.2×10^{-16} and K_a for HCN is 4.8×10^{-10} . The pH of the buffer is:

23. How many of the following compounds can exhibit acidic nature ?

$Be(OH)_2$, $Mg(OH)_2$, $Al(OH)_3$, $B(OH)_3$, $Si(OH)_4$, Ga_2O_3 , In_2O_3 , CO , CO_2 , GeO_2 , SnO_2 , PbO_2 , GeO , SnO , PbO , Li_2O , KO_2 , K_2O_2 .

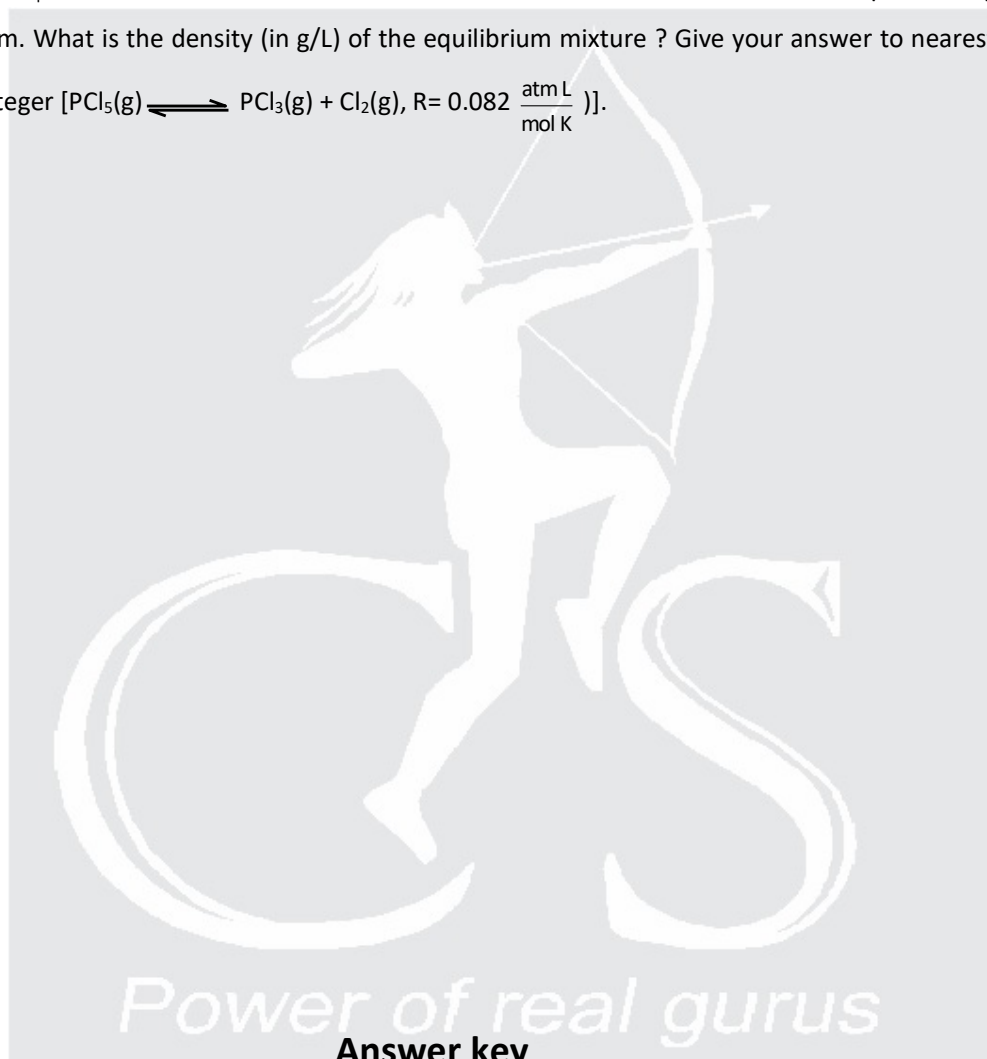
24. For the reaction $3A \longrightarrow 2B$, the following graph is obtained experimentally.



The rate of reaction when $[A] = 0.2 \text{ M}$ is $P \times 10^{-4} \text{ M min}^{-1}$. Value of $9P$ is -

25. The K_p value for the dissociation of 1 mol of PCl_5 at 500 K is 1.8 atm. The total equilibrium pressure is 1 atm. What is the density (in g/L) of the equilibrium mixture ? Give your answer to nearest single digit

integer $[\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}), R = 0.082 \frac{\text{atm L}}{\text{mol K}}]$.



Answer key

- | | | | | |
|------------|--|-----------|-----------|-----------|
| 1. (C) | 2. (D) | 3. (A) | 4. (B, D) | 5. (ABC) |
| 6. (ABC) | 7. (BC) | 8. (CD) | 9. (ABD) | 10. (BC) |
| 11. (ABCD) | 12. (AD) | 13. (ABD) | 14. (BCD) | 15. (AB) |
| 16. (ABD) | 17. (A) $\rightarrow$ R, (B) $\rightarrow$ P, (C) $\rightarrow$ Q, (D) $\rightarrow$ S | 18. (D) | | |
| 19. (A) | 20. (02.00) | 21. 06.00 | 22. 03.00 | 23. 12.00 |
| 24. 16.00 | 25. 03.00 | | | |

